

Hardware Project No. 4: Bridge Circuit for Transducer Interface

So, we know that in order to make physical measurements of things like force, we need to use a bridge circuit to interface the transducer with our electronics. So, how do we do this?

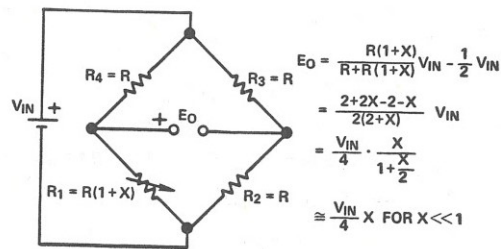


Figure 1: Wheatstone bridge circuit with transducer.

Figure 1 shows the basic Wheatstone bridge circuit, with a transducer in one arm of the bridge. When the bridge is “balanced” (i.e., all resistors are matched) the output voltage E_0 is 0. Ideally, the change in resistance due to the transducer is small; this will make the output sort of linear. So, first, one needs to construct a bridge with matched resistances (including the transducer) that balances. Additionally, the change in resistance from the transducer should not be too large. The gauge itself and matched resistors may be sufficient for this, or one may want to choose relatively large resistors for the arms of the bridge (large resistors will also limit current flow and recall that many things we are interested in interfacing with are temperature sensitive, so the less heat dissipated, the better). The other element of this circuit is the excitation voltage. Note that in real life, this has to be a very stable voltage, requiring use of specialized driver circuitry, including things like voltage regulators. Good thing you’ve got one! [The 7805 voltage regulator provided should supply a steady 5V output (check it to verify exactly what it is!) for input voltages between 7 and 25V]

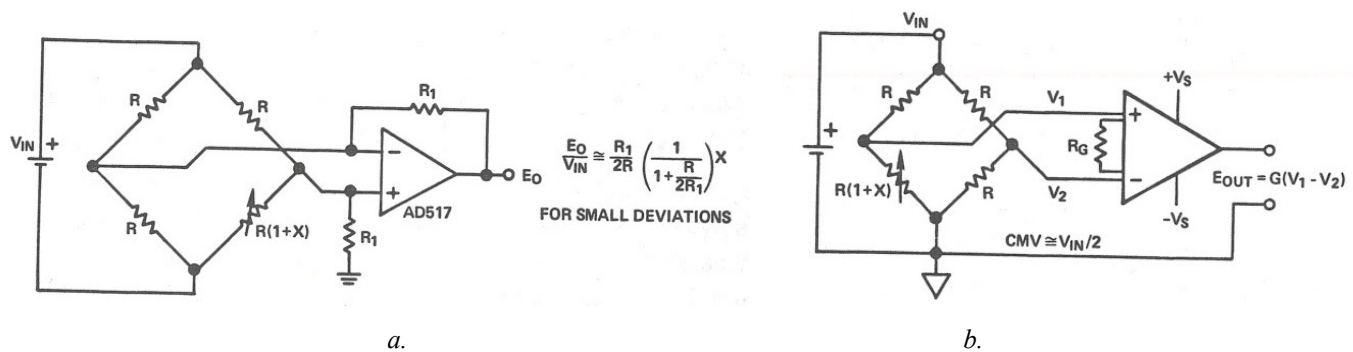


Figure 2: a. Single op-amp bridge amplifier. b. Instrumentation amp bridge amplifier.

Of course, if the bridge circuit and transducer generate very small voltages, one may desire amplification to make accurate measurements. How would one do that? Figure 2 shows two possibilities. (Note that Figure 2b assumes you are using an IC instrumentation amplifier, where the gain is set using a single external resistor, but you know how to build such things from scratch). How big should the gain of this amplifier be? The answer is: big enough. You want to be able to read something useful from it. Be aware that the power supplies for the amplifier should be separate from the excitation voltage for the bridge circuit. Also note: if the bridge circuit is setup such that it will only generate positive voltage changes, you will need only a positive power supply for any amplifier circuit.

That instrumentation amplifier could come in handy. What if, for example, you cannot quite match the resistors in the bridge so you get an output with no load applied? Recall that the basic instrumentation amp circuit includes an optional potentiometer to trim away any such artifact. Could be useful... Further (and importantly), the instrumentation amplifier setup provides isolation/buffering to the bridge circuit (it only connects the bridge outputs to op amp inputs; the circuit in 2a does not do such, so it could load the bridge circuit, which is something to consider).

Another variation is shown in Figure 3. This circuit conveniently minimizes any common-mode DC voltage, but requires a split excitation voltage for the bridge circuit.

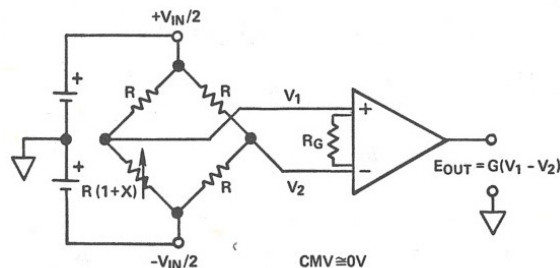


Figure 3: Minimization of common-mode voltage using center-tapped power supply for bridge excitation.

Your Task

Build a bridge circuit (with whatever excitation and amplification are required). Do you need amplification? Well, if you apply strain to the gauge, do you get millivolts output or a volt? Millivolts would call for some amplification. Recall also multiple gauge setups for increasing sensitivity.

How does one apply force? You could simply press down on the thing, or bend it. Or, you could set it up the way it would be in real life: attach it to something (a piece of plastic? Wood? Metal? Scotch taping it to a plastic knife or ruler or credit/ID card seems a good candidate...) and then try to bend the thing to which it is attached. I have a bunch of rulers that would be ideal for such purpose. The end of that cantilever you attach it to...you could conceivably use that to weigh things: do two quarters put on this thing produce twice the output voltage of one quarter? Do three quarters triple the output

voltage of a single quarter? What happens if you tap it and let it oscillate? With different weights attached? So, provide some representative outputs for various different loads you've applied.

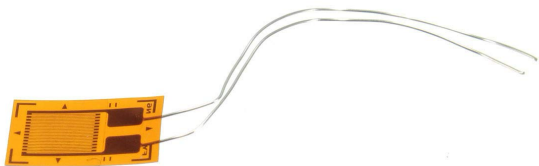
And, you have several gauges (and I have a few spares). Set this up and make measurements with a strain gauge. Then, add a second, opposite the first; does your sensitivity, in fact, double? Document with some measurements. Look at the notes from last week for how to properly mount them.

Bear in mind: those leads attached to the strain gauge? If they tug on the gauge, that is going to change your measurement (or they may just pull off completely). Be sure to supply some strain relief to the leads. And don't be surprised if you destroy one or two trying to solder longer lead wires.

Maybe [*you don't need to do any of this*, but it is, arguably, "fun"] you want to add some kind of display for something like this. Simple would be a few LEDs (say one lights up [or changes state] for a low force, two for a large force). You can hack something like that together: red LEDs drop about how much voltage across them? Or perhaps use different colors? Or maybe use a spiffy Arduino to display output if you find it simpler than a few LEDs. You could use any supporting circuitry you like for this, including transistors, op amps, digital logic, A/D converter chips, and the like. If you don't want to display, how about controlling something? Sound a buzzer if a certain force is applied? Wherever your imagination leads you, if you feel like controlling something that's not a few LEDs, it would be worth a shot. *Again, these are just ideas that you might get a kick out of doing, they are not a requirement.* **Feel free to entirely ignore this last paragraph.**

Due Date

April 23. Write a report describing your circuit and how it functions. And let's play with them in the lab.



Strain Gauge Specs [such as they are]

Resistance: $120 \pm 3 \Omega$

Gauge factor: $2.1 \pm 1 \%$ (gauge factor is the ratio of [change in resistance divided by resting resistance] and [change in length divided by resting length...that's strain, folks]:

$$GF = [\Delta R/R_0]/[\Delta L/L_0]$$

Strain limit: 2%

(You may want to check with a meter: is the thing actually 120Ω unloaded?)